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Assigning parcel destinations to drop-off points in a congested robotic sorting system

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Abstract

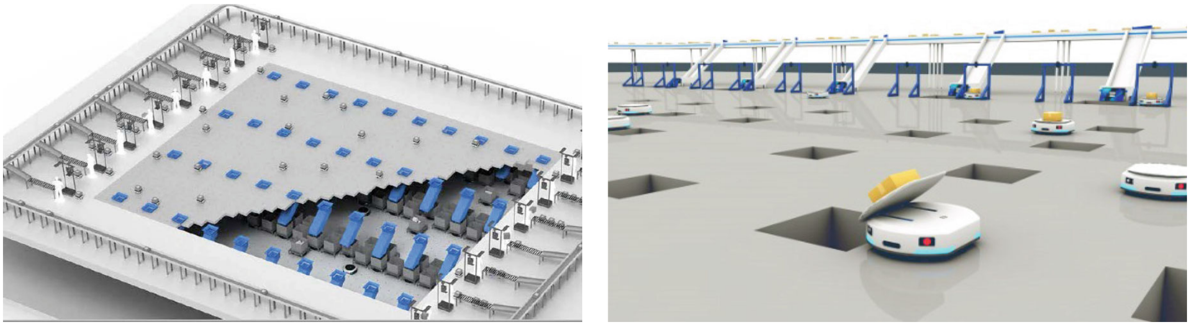
Autonomous mobile robots are increasingly used for order picking, order delivery, and parcel sorting. This article studies a robotic sorting system that uses robots to transport parcels from loading stations to drop-off points. While this system provides more flexible throughput capacity than conventional sorting systems, its performance is significantly affected by the robot travel distance and robot congestion. We study the problem of assigning parcel destinations to drop-off points to minimize the throughput time, trading off travel distance and congestion. First, an open queuing network (OQN) with finite capacity queues is constructed to estimate the congested throughput time. A decomposition method based on the analysis of the tandem queuing network of each aisle is developed to solve the OQN. Second, using the obtained throughput time as an objective and the destination assignments as decisions, we formulate an optimization model and solve the problem using an adaptive large neighborhood search (ALNS) algorithm. We validate the accuracy of the OQN by simulation and verify the efficiency of the ALNS algorithm by comparing it with Gurobi, a tabu search algorithm, several heuristic assignment rules, and the rule used by our case company, that assigns high demands close to loading stations. The results show that the ALNS solution provides a relatively low throughput time by dispersing destinations with high demands over drop-off points. In addition, we investigate the effects of different system layouts and travel path topologies. We also show that the ALNS assignment rule produces substantially lower operational costs than the heuristic assignment rules for a given required throughput capacity.

KEYWORDS

adaptive large neighborhood search, assignment problem, logistics, queuing network, robotic sorting system

1 | INTRODUCTION

Warehouses are increasingly using autonomous robots to improve throughput flexibility and lower operating costs in almost every aspect, such as order picking, inventory management, packaging, and parcel sorting. Robotic sorting systems (RSSs) have been introduced in recent years to handle sorting tasks in e-commerce and express companies, such as the Xanthus and Pegasus Fulfillment Robots of Amazon (Ames, 2019), and the sorting robots used by China Post, JD.com, Deppon, and Shentong Express (Zhang, 2019). Figure 1 presents a two-tier RSS. The top tier of the system is a mezzanine with many evenly distributed holes, each of which serves as a drop-off point (DP). Several loading stations are evenly distributed along the perimeter of the mezzanine in which parcels are placed onto a robot by a manual worker or an automated handler. A DP on the top tier is connected by a spiral conveyor that leads the parcels to a roll container installed on the bottom tier. A fleet of robots is used to transport parcels from the loading stations to the DPs and release them by a tilt mechanism (Figure 1B). The parcels for the same destination are collected at the bottom tier roll container for further delivery.



(A) (B)

FIGURE 1 Two-tier RSS.*Source*: Geek+.

Given the scalability of the robot fleet size and by opening or closing loading stations, RSSs enjoys a more flexible throughput capacity than does an automated conveyor-based sorting system. Moreover, the floor space needed for a robot sorting solution is much smaller than the needed for a sorting system using automated conveyors (Tompkins, 2021). However, RSSs face some challenges that may significantly affect the systems' performance. The first is robot congestion, which arises from a large number of robots operating within a limited space. This congestion directly reduces the system's throughput capacity by impeding the robots' movement and consequently increases the time required for parcels to be processed. The second is the assignment of parcel destinations to DPs, which determines the sorting demands among the DPs. This assignment can be adjusted easily in the control software during down times of the system. This situation affects both the robot travel distance and the robot congestion. However, a trade-off exists. Robot travel distance can be reduced by assigning high demands to DPs that are close to the loading station, a scheme that reduces the parcel throughput time. However, this rule aggravates robot congestion near these DPs, causing delays in the throughput time. Conversely, assigning low demands to the DPs that are close to the loading stations can alleviate robot congestion, but it increases the robot's travel distance. In practice, it is hard to predict the effect of an assignment rule on robot congestion, due to the lack of an insightful analytical model and fluctuating demands. This motivates us to study the assignment problem of parcel destinations to DPs. The following research questions are investigated in this article:

1. How can an analytical approach be developed to estimate the throughput time of an RSS while considering robot congestion?
2. How can parcel destinations be assigned to DPs to minimize the throughput time?
3. What is the effect of the destination demand, system layout, and path topology on the assignment of parcel destinations to DPs?

To answer the above questions, we first develop an OQN with finite capacity queues to estimate the throughput time that considers robot congestion. We decompose the tandem network of each aisle into multiple single queue systems, each with a finite buffer for waiting robots, to capture robot congestion. An iterative method is used to analyze the throughput rate of the tandem queuing network of each aisle. We show convergence in the case of exponential service distributions, while the general case is still open. On this basis, we transform the original OQN into a reduced OQN that has a product-form solution. This yields an estimate for the congested parcel throughput time. For general distributions, the solution method for the queuing network is approximative, and its performance is not guaranteed. Then, we build an optimization model for the destination assignment problem to minimize the throughput time obtained by the OQN. The model is proven to be NP-hard, and an ALNS algorithm is designed to solve the trade-off issue between the robot travel distance and robot congestion. We use simulation to validate the accuracy of the OQN and conduct numerical experiments to verify the effectiveness of the ALNS algorithm. The ALNS algorithm outperforms the truncated Gurobi solver in large instances. Then, we compare the ALNS assignment rule with both a tabu search algorithm and several heuristic rules: assigning high demands to DPs close to the loading station (HC), as used by our case company, and assigning low demands close to the loading station (LC). The ALNS evenly assigns destinations with high demands to DPs and; consequently, the throughput time is reduced by 2.3%, 17%, and 19% compared with that of the tabu search-based, HC, and LC rules, respectively. Motivated by this finding, we propose a balanced (BA) assignment rule that provides a throughput time 7.5% higher than that of ALNS. The ALNS algorithm also outperforms the HC rule used in our case company, leading to a 30% reduction in the company's throughput time, illustrating its practical superiority in managing throughput efficiency.

We also investigate (1) the effect of the destination demands, (2) the system layout (the width-to-length ratio), and (3) the robot travel path topology (by distinguishing aisles with single-lane or dual-lane traffic) on the throughput time and the assignment of destinations to DPs. The ALNS assignment rule is robust and outperforms the HC, LC, and BA rules in terms of throughput time. Although the layout of the system substantially affects the throughput time, the ALNS assignment rule is consistently the best. In addition, in a system with a compact topology (with unidirectional aisles), the ALNS assignment rule is preferred. Although using dual unidirectional lanes in aisles can alleviate robot congestion, the ALNS algorithm still disperses destinations with large demands among drop-off points when the system faces high demand. Finally, we examine the performance of the ALNS in terms of operational costs. Substantial cost savings can be achieved by using the ALNS in the system with a requirement on its throughput capacity, compared with the aforementioned heuristic assignment rules. These provide practical insights for the operating policies of RSSs, especially considering robot congestion.

This paper offers the following contributions. First, we examine the destination assignment problem in the RSS and design a queuing network that can accurately estimate the throughput time, considering robot congestion and various layouts and path topologies. Second, we design an ALNS algorithm to solve the destination assignment problem and comprehensively verify its effectiveness. Finally, we investigate how the assignment of destinations to DPs varies with destination demands, system layouts, and path topologies.

We proceed as follows: Section 2 reviews the literature on conventional sorting systems and robotic sorting systems. In Section 3, the performance estimation model is constructed and solved. In Section 4, we build an optimization model for the destination assignment problem in the RSS. In Section 5, we conduct numerical experiments to verify the performance of the ALNS assignment rule. In Section 6, we conclude this paper and provide several future research directions.

2 | LITERATURE REVIEW

This study is related to two streams of literature: studies on assignment problems in conventional sorting systems and studies on RSSs and assignment problems in other robotic warehousing systems. We first review the studies on the assignment problem of destinations to outbound stations in conventional sorting systems that resemble the destination assignment problem in RSSs. Then, we review the papers on RSSs and those focusing on assignment problems in other robotic warehousing systems, considering their operational similarity with RSSs.

Automated sorting systems typically have one or a few inbound stations (inducts) and multiple accumulation lanes (outputs). Existing studies on the assignment problems of such systems examine either the assignment of incoming orders to inbound stations or outbound destinations to accumulation lanes (Boysen et al., 2019). We focus on the papers dedicated to the assignment of outbound destinations to accumulation lanes. Meller (1997) optimized the assignment problem of incoming orders to outbound lanes in an order accumulation/sortation system with a fixed arrival sequence of items. He formulated a mixed-integer programme with the minimization of the maximum time an order recirculating on the conveyor. The problem was solved by a decomposition approach. Johnson (1998) examined the inbound orders to outbound lane assignment of a generic sortation system consisting of a single induction lane, a recirculation conveyor, and multiple outbound accumulation lanes. The result showed that assigning the next available order to an unassigned accumulation lane could outperform any fixed lane assignment rule in terms of throughput time.

Incoming shipments must also be sorted in a cross-docking terminal, in which products received from inbound trucks are immediately unloaded, sorted, consolidated, and loaded into outbound trucks for delivery with little or no storage in between (Rijal et al., 2019). The assignment of inbound and outbound destinations to inbound and outbound dock doors (called the dock-door assignment problem) is a major and challenging operational decision at a cross-docking terminal. Depending on whether there are more destinations than dock doors, either a static or a dynamic dock-door assignment policy may be used (Boysen et al., 2019). Several studies have investigated the congestion in the static dock-door assignment problem. Gue (1999) studied the assignment problem of inbound doors to incoming trucks at a cross-docking center for freight handling (also referred to as less-than-truckload), to minimize the internal travel distances of workers. A case study showed that the look-ahead policy could lead to 15% to 20% lower labor costs than a first-come-first-serve policy. Bartholdi and Gue (2000) considered the congestion caused by interference among forklifts, and further constraints in the travel cost model of Gue (1999). They used a steady-state queuing model to estimate the average waiting time resulting from the interference among forklifts. Bozer and Carlo (2008) investigated the assignment of inbound/outbound trailers to dock doors in a less-than-truckload cross-docking center while aiming at minimizing the total internal travel distance.

If there are more destinations than dock doors in a cross-docking terminal, then dynamic dock-door assignment is needed to schedule trucks and assign destinations to dock doors. Chen and Lee (2009) addressed the dynamic assignment of inbound/outbound trucks to two dedicated doors by formulating this problem as a two-machine flow shop problem. They proved the NP-hardness of this problem and design a heuristic for the solution. Chen and Song (2009) generalized the problem by modeling multiple inbound and outbound gates by parallel machines. Miao et al. (2009) considered the dynamic assignment problem of trucks to doors in a cross-docking terminal with an operational time constraint and took the minimization of the operational cost of the cargo shipments and the total number of unfulfilled shipments as the objective. They formulated this problem as an integer programming model and designed a tabu search and a genetic algorithm for the solution. Building on this foundation, subsequent work (Miao et al., 2014) has introduced an adaptive tabu search algorithm, further refining the approach to study the truck-to-dock assignment problem in the cross-docking terminal. Other models with promising computational performance have also been formulated for this problem (Gelareh et al., 2020; Nassief et al., 2018). In addition, the dock door assignment problem is often jointly optimized with the truck scheduling problem, but the minimization of the total weighted tardiness (Liao et al., 2013), the total operation time (makespan) (Kuo, 2013), the maximum inventory (Zenker & Boysen, 2017), the operational costs (Enderer et al., 2017; Essghaier et al., 2021; Rijal et al., 2019).

The most relevant studies are those about assignment problems in robotic warehousing systems. Several papers consider the robotic sorting system. Zou et al. (2021) built closed queuing networks for the performance estimation of both single-tier and two-tier RSSs. They investigated the problem of assigning robots to loading stations, considering the random, dedicated, closest, and shortest-queue assignment rules. They ignored robot congestion in the analytical model of two-tier RSSs and used a traffic flow function to approximate robot congestion in single-tier RSSs. The paper that is tightly related to this study is that of Xu et al. (2022), who examined the distribution of arriving parcels over loading stations in RSSs. They built an open queuing network to estimate the throughput time and developed an optimization model with the objective of minimizing the throughput time. A tabu search algorithm was designed to solve the optimization model. The problem examined by Xu et al. (2022) traded off the expected waiting time of parcels at loading stations and the expected travel time of robots from loading stations to drop-off points. However, our problem aims to balance robot congestion and robot travel distance. To achieve this aim, our analytical model should be able to assess the effect of robot congestion on the mezzanine on the throughput time. Shi et al. (2022) considered a human–robot hybrid sorting system that used Internet-of-Things sensor technology to capture information on incoming parcels. They first built a regression model to predict the congestion of robotic sorters using the information captured by the Internet-of-Things and designed an online algorithm to solve the allocation problem of parcels between humans and robotic sorters. Boysen et al. (2023) optimized the assignment of SKUs to orders, orders to collection points, and transport jobs to robots in an RSS to minimize the makespan of a given arrival sequence of SKUs. The assignment of robots or orders to workstations has also been studied in robotic mobile fulfillment systems (Boysen et al., 2017; Shi et al., 2021; Tan et al., 2021; Wang et al., 2022; Xie et al., 2021; Zhuang et al., 2022; Zou et al., 2017). These studies generally built optimization models under deterministic demands, without considering robot congestion. Although two cross-docking studies (Bartholdi & Gue, 2000; Bozer & Carlo, 2008) took congestion, they merely examined congestion at a single point by estimating the waiting time at dock doors.

Our study differs from the previous works in two aspects. First, the assignment problem investigated in this paper affects both the robot travel distance and robot congestion on the mezzanine of the RSSs. The assignment problems examined in previous studies only affect the working efficiency of either workers or robots. Second, the previous studies generally considered deterministic demands (Boysen et al., 2017; Boysen et al., 2023; Tan et al., 2021; Xie et al., 2021). We estimate robot congestion under stochastic demands. An exception is the paper of Shi et al. (2022), who also studied congestion. However, their regression model did not investigate the effect of system layout, path topology, and most importantly, the assignment of destinations to drop-off points on the system throughput time. We summarize the key studies in Table 1 to show the differences between our paper and other papers studying destination assignment problems in internal transport systems. We not only investigate the assignment of parcel destinations to DPs in RSSs but also consider robot congestion in the travel path. Congestion is typically ignored in papers studying sorting or robotic warehousing systems, but it cannot be ignored in RSSs in which numerous robots operate within a confined space.

3 | THROUGHPUT TIME ESTIMATION OF CONGESTED RSSS

In this section, we build an OQN to estimate the throughput time of the RSS with a given assignment of destinations to drop-off points. This aspect provides the objective function of the optimization model in Section 4. We begin with a detailed description of the RSS, followed by the fundamental assumptions and primary notations. Then, we describe the formulation of the OQN, specify the solution approach, and present the accuracy validation results of the OQN.

3.1 | System description and modeling preparations

Figure 2 shows the top view of the RSS with n loading stations located along the north side of the system and $L \times W$ DPs (denoted as $d_{i,j}, i = 1, \dots, L, j = 1, \dots, W$) arranged in the central area of the mezzanine. Both aisles (denoted as a_j) and cross-aisles (denoted as ca_i) are evenly distributed on the mezzanine for robot travel. Throughout the paper, we use l to index a loading station, i to index a cross-aisle, and j to index an aisle. A drop-off point $d_{i,j}$ is indexed by both its neighboring cross-aisle ca_i and aisle a_j. We use $[\![N]\!]$ to denote the set of integers ranging from 1 to N, that is, $[\![N]\!] = 1, \dots, N$. Then we define the set of all DPs as $\mathcal{D} = \{d_{i,j} | i \in [\![L]\!], j \in [\![W]\!]\}$ and the set of all loading stations as $\mathcal{LS} = [\![n]\!]$. The parcel is first sorted by the conveyor to a loading station and transferred by the worker onto a robot. The robot finally transports the parcel to the designated drop-off point and releases it. We take the task from loading station n to drop-off point $d_{3,4}$ as an example to describe the robot path. The robot first travels west-to-east in the cross-aisle ca_0 to the entrance of aisle a_3. Then, the robot takes a 90-degree turn and travels to $d_{3,4}$ in a_3 (blue dashed line). After the robot releases the parcel into

TABLE 1 Summary of the literature on assignment problems in conveyor-based sorting systems and robotic sorting systems.

Reference	System	Congestion	Assignment problem	Metric
Meller (1997), Johnson (1998)	Sorting system	×	Orders to accumulation lanes	Throughput time
Tsui and Chang (1990, 1992), Gue (1999)	Cross-docking	×	Destinations to dock doors	Travel distance
Bartholdi and Gue (2000)	Cross-docking	✓	Destinations to dock doors	Travel cost
Bozer and Carlo (2008)	Cross-docking	✓	Destinations to dock doors	Workload
Chen and Lee (2009), Chen and Song (2009), Liao et al. (2013), Kuo (2013)	Cross-docking	×	Destinations to dock doors	Makespan
Miao et al. (2009), Miao et al. (2014), Enderer et al. (2017), Nassief et al. (2018), Rijal et al. (2019), Gelareh et al. (2020), Essghaier et al. (2021)	Cross-docking	×	Destinations to dock doors	Operational cost
Zenker and Boysen (2017)	Cross-docking	×	Destinations to dock doors	Maximum inventory
Zou et al. (2021)	RSS	✓	Robots to loading stations	Throughput capacity & cost
Xu et al. (2022)	RSS	✓	Parcels to loading stations	Throughput time
Shi et al. (2022)	RSS	✓	Assignment of parcels to human and robotic sorters	Cost
Boysen et al. (2023)	RSS	×	Piece-to-order, order-to-collection point, transport job-to-robot	Throughput time
This study	RSS	✓	Parcel destinations to DPs	Throughput time & cost

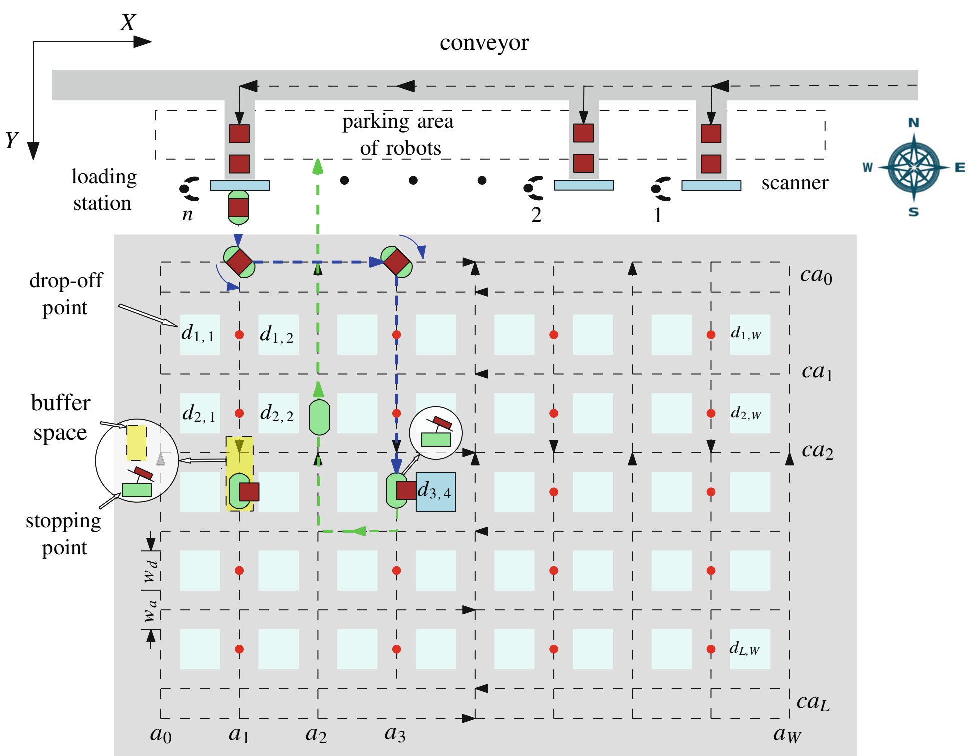


FIGURE 2 Layout of the RSS.

$d_{3,4}$, it travels back to the parking area of the robot following the green line. Specifically, the robot first transits through the first cross-aisle it meets (ca_3) to the aisle (a_2), then travels to the parking area.

Each aisle a_j and cross-aisle ca_i (except for ca_0 and ca_L) in Figure 2 has a unidirectional lane (black dashed line with an arrow) that specifies the traveling direction of the robot in this topology. Such a path topology leads to a compact mezzanine in which robot congestion is more significant than in a system with more lanes in each aisle and cross-aisle. We use this topology to illustrate the formulation of the performance estimation model in this section and later extend the model to a path topology with dual lanes (bidirectional travel) in aisles and cross-aisles (see Figure A1 in Appendix 1). The design trade-off between different path topologies was investigated in Zou et al. (2021). Another aspect to be clarified is that a robot releases the parcel in a north-to-south aisle, and returns to a loading station through a south-to-north aisle. While this setting limits the optimization of robot routing, which is beyond the scope of this study, it is in line with the practice of the RSS (Tompkins, 2021). We define the set of all north-to-south aisles as $\mathcal{A} = \{a_j | j \in [\![W]\!]'\}$, where $[\![W]\!]'$ is the set of odd integers ranging from 1 to W, that is, $j = 1, 3, \ldots, W-1$ (assuming W is an even number). In each aisle $a_j \in \mathcal{A}$, the robot can stop at specific points, termed stopping points $s_{i,j}$ (denoted by the red points in Figure 2), and the set of stopping points is $\mathcal{S} = \{s_{i,j} | i \in [\![L]\!], j \in [\![W]\!]'\}$. For a designated drop-off point $d_{i,j}$, if the index j is odd, then the corresponding stopping point $s_{i,j}$ is situated on the right side; conversely, if j is even, then the stopping point is positioned on the left side.

A robot may experience congestion in both the path from loading stations to DPs (called the forward path) and the path from DPs back to loading stations (the backward path). We only take the former into account, based on two considerations. On the one hand, we aim to assign parcel destinations to DPs to minimize the expected throughput time a parcel experiences on the mezzanine, which is mainly related to the robot traveling process in the forward path. On the other hand, each robot will stop to release the parcel in the forward path, which may produce contiguous congestion, while it will not actively stop in the backward path unless it is blocked. The robots operate under the blocking-after-service mechanism, as detailed by Akyildiz and Perros (1989). According to this mechanism, a robot being blocked will wait at its position until its target position, such as a stopping point, becomes accessible. Each stopping point $s_{i,j}$ can be occupied by one robot only and allows at most $C_{i,j}$ additional robots waiting at buffer $B_{i,j}$, adjacent to $s_{i,j}$, with , $s_{i,j} \in \mathcal{S}$. The foundational assumptions of our model are outlined as follows:

1. Parcels assigned to drop-off point $d_{i,j}, d_{i,j} \in \mathcal{D}$ arrive according to a Poisson process with an arrival rate $\lambda_{i,j}$. Other arrival processes can also be examined in the performance estimation model, without changing the model or the solution approach.
2. The time for a worker at the loading station to put a parcel onto a robot follows a uniform distribution $U[\underline{t}_{load}, \bar{t}_{load}]$. Another distribution can be used by adjusting the calculation of the first two moments of time (Equation (2)).
3. Robot congestion in cross-aisle ca_0 is not considered because it generally has multiple parallel lanes for robot traveling (dual lanes in Figure 2), and the robots will not stop in it to release parcels. As a result, a robot can travel at a nominal constant speed in cross-aisle 0. This scenario in line with both the practice and literature (Xu et al., 2022; Zou et al., 2021).
4. We ignore the acceleration and deceleration of robots. Accelerating to full speed takes very little time. Moreover, this assumption is common in the literature on robotic warehousing systems (Fragapane et al., 2021; Lamballais et al., 2017; Roy et al., 2012; Roy et al., 2020).
5. Sufficient robots are used in the system. The number of robots is determined at the design phase to meet peak demand, which guarantees their availability in daily operations. The loading stations are typically the system bottleneck (Sheu & Choi, 2023) because employing fewer workers is more cost-efficient than using fewer robots.

The formulation of the performance estimation model uses additional system parameters, including the time for a robot/worker to release a parcel (t_{lu}), the time for a robot to take a 90-degree turn (t_t), the conveyor maximum velocity (v_c), the robot maximum velocity (v_m), and the width of an aisle/cross-aisle (w_a) and a DP (w_d) (Figure 2). The values of these parameters are based on the work of Xu et al. (2022).

3.2 | Performance estimation model of the RSSs

We build an OQN model that can accurately estimate the (congested) throughput time resulting from a given assignment of parcel destinations to drop-off points of the RSS. System throughput time is popularly used in various robotic warehousing systems to evaluate configuration design and perform operating policy analysis. (Azadeh et al., 2019; Fragapane et al., 2021; Lamballais et al., 2020; Roy et al., 2020; Shi et al., 2022). We choose an OQN with finite queue buffers given the relative simplicity of modeling congestion and utilize the decomposition method outlined by Dallery and Frein (1993) and the method from Whitt (1983b) to obtain the performance metrics of the RSS.

Figure 3 shows the OQN. Parcels arrive according to a Poisson process with an arrival rate λ (per hour), which is the sum of the arrival rates to all drop-off points, that is, $\lambda = \sum_{i\in[\![L]\!]} \sum_{j\in[\![W]\!]} \lambda_{i,j}$. Parcels are transported by a conveyor

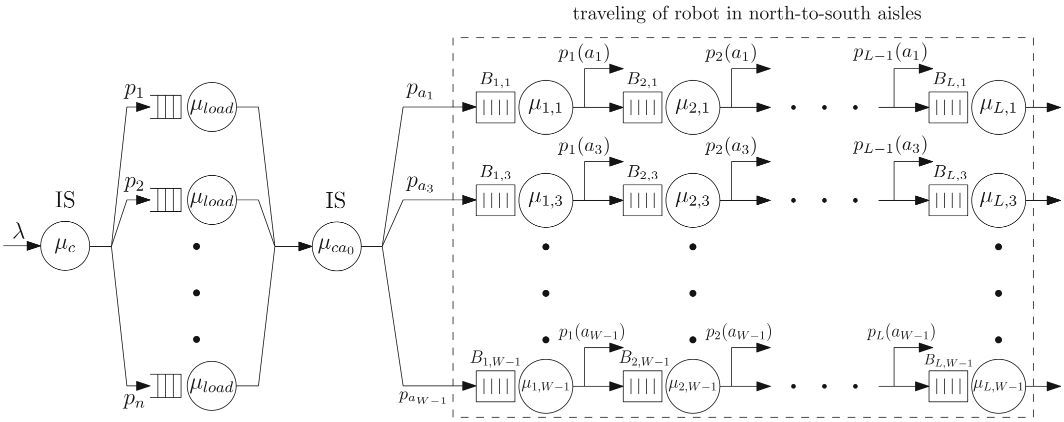


FIGURE 3 OQN for throughput time estimation of congested RSSs.

to the loading stations, which is modeled as an infinite service (IS) node with service rate μ_c. Although parcels may wait at the inbound doors before they are offloaded on the conveyor (not in the scope of this paper), once on the conveyor, they are directly transported to loading stations without additional waiting, so an infinite-server (i.e., a delay) model is appropriate. To balance the workload among workers at loading stations, parcels are assigned to all loading stations with equal probability $p_l = \frac{1}{n}, l \in \mathcal{LS}$. The worker at the loading station puts a parcel onto a robot, which is modeled as a single server with service rate μ_{load}. Then, the robot moves in the cross-aisle ca_0 to the north-to-south aisle neighboring the designated drop-off point. The robot movement in ca_0 is modeled as an infinite server with service rate μ_{ca_0} following the third assumption specified in Section 3.1. We define the probability that the robot would proceed to aisle a_j for parcel release as $p_{a_j}, a_j \in \mathcal{A}$. This probability is calculated as the ratio of the aggregate demand of drop-off points near a_j to the total demand λ. Therefore, $p_{a_j} = \sum_{i \in [\![L]\!]} (\lambda_{i,j} + \lambda_{i,j+1})/\lambda$ for any $j \in [\![W]\!]'$, and we have $\sum_{a_j \in \mathcal{A}} p_{a_j} = 1$.

The traveling of a robot in the north-to-south aisle $a_j \in \mathcal{A}$ is modeled as a tandem queuing network (the part in the dashed rectangle), which is a serial connection of L single service nodes (denoted as $\mu_{i,j}, i \in [\![L]\!]$). Each node has a finite queue buffer $B_{i,j}$ to accommodate the robot that attempts to visit the stopping point $s_{i,j}$. Once the robot occupies $s_{i,j}$, it either stops releasing the parcel into a neighboring drop-off point ($d_{i,j}$ or $d_{i,j+1}$) with a conditional probability $p_i(a_j)$ or traverses the point with a probability $p_i'(a_j) = 1 - p_i(a_j)$. After a drop, the robot immediately returns to the parking area via a cross-aisle (Figure 2). The service times of these two cases are integrated into the service rate $\mu_{i,j}$, which will be specified in the next section. We only present $p_i(a_j)$ in Figure 3 for clarity. It equals the ratio between the demand of drop-off points neighboring the stopping point $s_{i,j}$, that is, $\lambda_{i,j} + \lambda_{i,j+1}$, and the sum of demands of all the drop-off points the robot may traverse after leaving $s_{i-1,j}$. Therefore, we have $p_i(a_j) = (\lambda_{i,j} + \lambda_{i,j+1}) / \sum_{i'=i}^{L} (\lambda_{i',j} + \lambda_{i',j+1})$. For each aisle $a_j \in \mathcal{A}$, we have $p_L(a_j) = 1$ because the robot will certainly release the parcel into either $d_{L,j}$ or $d_{L,j+1}$ when it arrives at the stopping point $s_{L,j}$, and then leaves a_j.

The OQN in Figure 3 differs from previous performance estimation models of RSSs (Xu et al., 2022; Zou et al., 2021) on two aspects. First, our OQN specifically focuses on the robot congestion in aisles by modeling each stopping point as a node with finite buffer capacity, while previous models either investigate the effect of robot congestion on the system throughput capacity by simulation (Zou et al., 2021) or capture the robot congestion on the conveyor that sorts parcels to loading stations (Xu et al., 2022). Our method is based on the consideration that robot congestion is an important factor in the assignment of destinations to DPs in the RSS. Second, the robot congestion at a stopping point in the aisle depends on the probability that the robot would release the parcel at the stopping point in the aisle, that is, $p_i(a_j)$. This scenario makes both the service time and the structure of the tandem OQN of the aisle different from the model in the paper of Xu et al. (2022). As a result, a decomposition method is designed in the next section to solve the OQN. While this approach is an approximation method that may produce some errors, we validate its accuracy with a simulation model. The numerical results show that the robot congestion significantly affects the throughput time.

3.3 | Performance analysis of OQN

The OQN does not have a product-form solution (Bolch, 2006), due to the finite buffer capacity of the service nodes $\mu_{i,j}, i \in [\![L]\!], j \in [\![W]\!]'$. To solve this problem, we design a decomposition method which is a popularly used method for analyzing complicated nonproduct-form queuing networks, typically leading to somewhat accurate results for systems with not too low utilization (Dallery & Frein, 1993; Kuehn, 1979; Morabito et al., 2014; Rabta, 2009). The method involves a series of procedures. 1) Analyze the tandem open queuing network of each north-to-south aisle by the two-server decomposition method for a tandem open queuing network with finite queue buffers (Dallery & Frein, 1993).

On this basis, we obtain the mean throughput rate (expected number of parcels being processed in unit time) of aisle $a_j \in \mathcal{A}$, denoted as TR_{a_j}. 2) Replace the queuing system of aisle a_j by a delay node μ_{a_j} with service rate $\mu_{a_j} = TR_{a_j}$. 3) Transform the original OQN into a reduced OQN that can be solved by decomposing it into several independent server nodes and calculating their performance metrics (Marchet et al., 2012).

3.3.1 | Calculation of service time of nodes in the OQN

We first calculate both the mean and squared coefficient of variation (scv) of the service times of each queuing node in the OQN to serve as inputs for the above calculation process. The mean and scv of the movement time of parcels on the conveyor can be calculated as follows:

$$\mu_c^{-1} = \sum_{l\in\mathcal{LS}} p_l\left(\frac{W(w_a+w_d)-X_l}{v_c}\right),$$
$$cv_c^2 = \frac{\sum_{l\in\mathcal{LS}} p_l\left(\frac{W(w_a+w_d)-X_l}{v_c}\right)^2}{\mu_c^{-2}} - 1, \tag{1}$$

where X_l is the known X-axis coordinate of loading station l for $l \in \mathcal{LS}$.

The operating time of a worker is assumed to follow a uniform distribution $U[\underline{t}_{load}, \bar{t}_{load}]$, as detailed in assumption 2. The mean and scv of the service time at a loading station can be obtained by

$$\mu_{load}^{-1} = \frac{\underline{t}_{load} + \bar{t}_{load}}{2}, \quad cv_{load}^2 = \frac{(\bar{t}_{load} - \underline{t}_{load})^2}{3(\underline{t}_{load} + \bar{t}_{load})^2}. \tag{2}$$

The service time at node μ_{ca_0} consists of three parts. 1) The robot takes a 90-degree turn to enter cross-aisle ca_0 after leaving a loading station. 2) The robot leaves loading station l with probability p_l, and travels a distance $|X_{a_j} - X_l|$ to aisle a_j with probability p_{a_j}, where X_{a_j} is the X-axis coordinate of aisle a_j, and is calculated by $X_{a_j} = j \cdot (w_a + w_d)$. 3) The robot takes a 90-degree turn to enter aisle a_j. The first and third parts take time $2t_t$, and the second part takes time $\frac{|X_{a_j}-X_l|}{v_m}$ with probability $p_l \cdot p_{a_j}$. Therefore, we calculate the mean and scv of the movement time of robots in cross-aisle ca_0 by

$$\mu_{ca_0}^{-1} = \sum_{l\in\mathcal{LS}}\sum_{a_j\in\mathcal{A}} p_l p_{a_j}\left(\frac{\left|X_{a_j}-X_l\right|}{v_m} + 2t_t\right),$$
$$cv_{ca_0}^2 = \frac{\sum_{l\in\mathcal{LS}}\sum_{a_j\in\mathcal{A}} p_l p_{a_j}\left(\frac{\left|X_{a_j}-X_l\right|}{v_m} + 2t_t\right)^2}{\mu_{ca_0}^{-2}} - 1. \tag{3}$$

The service of a robot at service node $\mu_{i,j}, i \in [\![L]\!], j \in [\![W]\!]'$ has two options. 1) The robot travels a distance of w_a+w_d without stopping. 2) The robot stops to release the parcel and takes a 90-degree turn to leave aisle a_j. The first and second options happen with the probability $p_i'(a_j)$ and $p_i(a_j)$, respectively. Based on these options, the mean and scv of the service time at service node $\mu_{i,j}$ can be calculated as follows:

$$\mu_{i,j}^{-1} = p_i'(a_j)\left(\frac{w_a+w_d}{v_m}\right) + p_i(a_j)\left(\frac{w_a+w_d}{v_m} + t_{lu} + t_t\right),$$
$$cv_{i,j}^2 = \frac{p_i'(a_j)\left(\frac{w_a+w_d}{v_m}\right)^2 + p_i(a_j)\left(\frac{w_a+w_d}{v_m} + t_{lu} + t_t\right)^2}{\mu_{i,j}^{-2}} - 1. \tag{4}$$

3.3.2 | Throughput analysis of the tandem queuing network of each north-to-south aisle

We analyze the throughput rate of the tandem queuing network of each north-to-south aisle $a_j \in \mathcal{A}$ using the decomposition method for tandem queuing networks with blocking, as proposed by Dallery and Frein (1993). This method decomposes the tandem queuing network into multiple subsystems, each consisting of two adjacent service nodes connected by an intermediate buffer. To obtain the service rate of the tandem queuing network, an iterative algorithm is proposed, of which the convergence is guaranteed in Proposition 1.

Proposition 1. *The iterative algorithm for solving the tandem queuing network of aisle $a_j \in \mathcal{A}$ converges for stable systems, when the service times of both the upstream and downstream servers of each subsystem follow exponential distributions.*

Proof. Refer to Appendix 4. ■

The assumption that the service time of each subsystem is exponentially distributed is restrictive. We therefore use a general distribution to model the inter-arrival and service times of each service node within the tandem queuing network of each aisle a_j. Using the first two moments of the inter-arrival times and service times, the model can capture the variability of service time, thereby enhancing its applicability to real-world scenarios. Each subsystem is assumed to be stable with independent arrivals. This simplifies the analysis but may not accurately reflect the interactions in the real-world system. Although convergence of the iterative algorithm cannot be demonstrated in the general case, it converges rapidly in all instances tested. This approach is supported by the numerical results obtained by Manitz (2015). In addition, the analytical model is validated with discrete event simulation, as described in Section 3.4. The error in throughput time appears to be less than 3%.

For clarity, we describe the procedure below and include the details in Appendix 3. Figure 4 shows the decomposition of the tandem queuing network of aisle a_j. A service node $\mu_{0,j}$ is first added to represent the arrival process of the aisle a_j. The mean service rate of this node equals the inter-arrival rate of this aisle, that is, $\mu_{0,j} = \lambda_{a_j}$. The scv of the service time of this node represents the scv of the inter-arrival time of robots to this aisle. This variable can be obtained

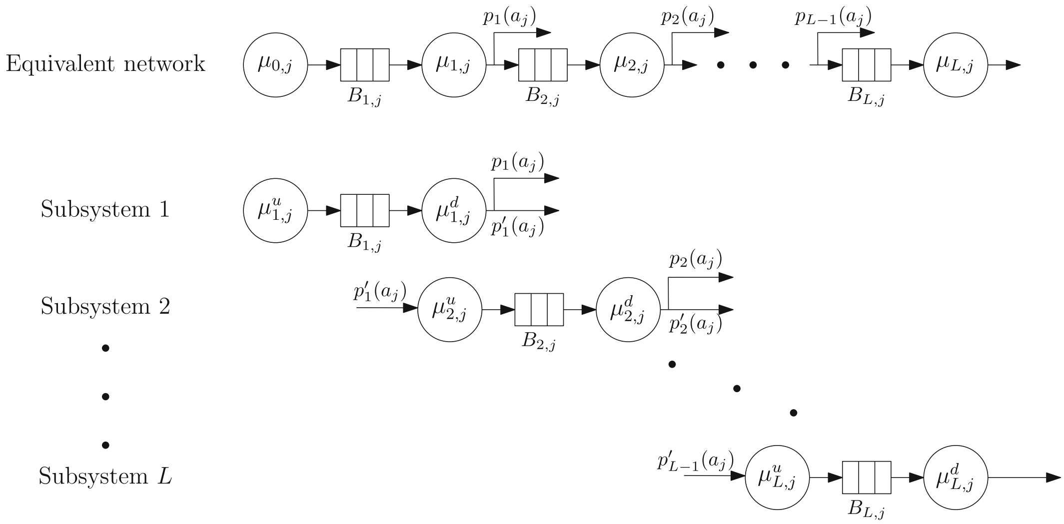


FIGURE 4 Decomposition of the tandem queuing network of aisle a_j.

by analyzing the inter-departure rate of node μ_{ca_0}, which is $cv^2_{a_j,A} = \rho^2_w\left(cv^2_{w,B} - 1\right)$ (Pujolle and Ai, 1986; Whitt, 1983a; Whitt, 1983c, refer to Appendix 2 for details).

Each subsystem consists of an upstream server with service rate $\mu^u_{i,j}$, a finite buffer $B_{i,j}$, and a downstream server with service rate $\mu^d_{i,j}$. The upstream server represents the arrival process of robots into the buffer, and its effective service time consists of a starvation time and a service time at the preceding subsystem $i-1$. Starvation occurs when buffer $B_{i-1,j}$ is empty and persists until service completion at subsystem $i-1$. The completion-instant starvation probability of subsystem $i-1$ in aisle j is defined as $p^s_{i-1,j}$. The starvation time of subsystem i can be approximated by the remaining service time of the upstream server of subsystem $i-1$ when buffer $B_{i-1,j}$ is empty. The effective service time of the upstream server $\mu^u_{i,j}$ is calculated as follows:

$$\frac{1}{\mu^u_{i,j}} = p'_{i-1}(a_j)\cdot\left(\frac{1}{\mu_{i-1,j}} + \frac{p^s_{i-1,j}}{\mu^u_{i-1,j}}\right), \quad i = 2, \dots, L, \tag{5}$$

where $p'_{i-1}\left(a_j\right)$ denotes the probability of robots leaving subsystem $i-1$ and going to its subsequent subsystem i within aisle a_j. Using the approximation of Manitz (2015), the *scv* of the effective service time of the upstream server $\mu^u_{i,j}$ is derived as follows:

$$cv^2_A(i,j) = (\mu^u_{i,j})^2\left(\frac{cv^2_{i,j}+1}{\mu^2_{i,j}} + p^s_{i-1,j} \cdot\left(\frac{cv^2_A(i-1,j)+1}{(\mu^u_{i-1,j})^2} + \frac{2}{\mu_{i,j}\cdot\mu^u_{i-1,j}}\right)\right), \quad i = 2, \dots, L. \tag{6}$$

Similarly, the downstream server $\mu^d_{i,j}$ represents the leaving process of robots from node $\mu_{i,j}$. The effective service time of this server consists of the remaining service time at node $\mu_{i,j}$ and the blocking time at subsystem i. Blocking occurs when buffer $B_{i+1,j}$ is full and persists until the service completion of node $\mu_{i+1,j}$. We define the completion-instant probability of buffer $B_{i+1,j}$ being full as $p^b_{i+1,j}$. Then, we calculate the blocking time of subsystem i as the remaining service time of node $\mu_{i+1,j}$ on the condition of buffer $B_{i+1,j}$ being full. The effective service time of the downstream server $\mu^d_{i,j}$ is given by

$$\frac{1}{\mu^d_{i,j}} = \frac{1}{\mu_{i,j}} + \frac{p^b_{i+1,j}}{\mu^d_{i+1,j}}, \quad i = 1, \dots, L-1. \tag{7}$$

The *scv* of the service time of the downstream server $\mu^d_{i,j}$ is given by

$$cv^2_D(i,j) = (\mu^d_{i,j})^2\left(\frac{cv^2_{i,j}+1}{\mu^2_{i,j}} + p^b_{i+1,j} \cdot\left(\frac{cv^2_D(i+1,j)+1}{(\mu^d_{i+1,j})^2} + \frac{2}{\mu_{i,j}\cdot\mu^d_{i+1,j}}\right)\right), \quad i = 1, \dots, L-1, \tag{8}$$

where $p^s_{i,j}$ and $p^b_{i,j}$ represent completion-instant probabilities. These variables are derived using the steady-state probability $p(k,i,j)$ of subsystem i (k customers in node $\mu_{i,j}$, $k \le C_{i,j}+1$), which are presented as follows (Dallery & Frein, 1993):

$$p^b_{i,j} = \frac{p(C_{i,j},i,j)}{1-p(C_{i,j}+1,i,j)}, \quad p^s_{i,j} = \frac{p(1,i,j)}{1-p(0,i,j)}, \tag{9}$$

where the steady-state probability $p(k,i,j)$ is obtained using the method proposed by Buzacott and Shanthikumar (1993) (refer to Appendix 5 for further details).

Based on Equations (5), (6), (7), (8), and (9), we use forward iteration to first calculate the values of the completion-instant starvation probability $p^s_{i,j}$, service rate $\mu^u_{i,j}$, and $cv^2_A(i,j)$ of the upstream servers. Then, we iterate backward to calculate the values of the completion-instant blocking probability $p^b_{i,j}$, service rate $\mu^d_{i,j}$, and $cv^2_D(i,j)$ of the downstream servers. Repeat the forward and backward iteration unless the changes of both $\mu^u_{i,j}$ and $\mu^d_{i,j}$ are within a given threshold, and the algorithm terminates. Details can be found in Algorithm 1 in Appendix

Algorithm 1. Balanced assignment rule of destinations to DPs in RSSs.

Input: The original sets of destinations $DS = \{ds_1, \dots, ds_m\}$ and DPs $DP = \{dp_1, \dots, dp_m\}$.
Sort the destinations in DS by demand in ascending order and obtain the sorted set, denoted as $DS' = (ds'_1, \dots, ds'_m)$.
Divide the sorted set DS' into $\frac{m}{2}$ subsets, denoted as $Sub_i = (ds'_i, ds'_{m-i+1}), i = 1, \dots, \frac{m}{2}$.
Let $k = 1$
for $i = \frac{m}{2}, \frac{m}{2} - 2, \dots, 2$ **do**
  Assign the destinations in Sub_i to the DPs dp_k, dp_{k+1}.
  Let $k = k + 2$.
end for
Let $k = 1$
for $i = 1, 3, \dots, \frac{m}{2} - 1$ **do**
  Assign the destinations in Sub_i to the DPs $dp_{\frac{m}{2}+k}, dp_{\frac{m}{2}+k+1}$.
  Let $k = k + 2$.
end for

3. Finally, we calculate the expected throughput rate of aisle a_j as

$$TR_{a_j} = \sum_{i \in [\![L]\!]} p_i(a_j) \cdot \mu^d_{i,j}, \quad j \in [\![W]\!]', \tag{10}$$

where $p_i(a_j) \cdot \mu^d_{i,j}$ indicates the rate of a robot leaving aisle a_j after traversing stopping point $s_{i,j}$.

3.3.3 | Throughput analysis of the reduced OQN of the RSS

We replace the tandem queuing system of aisle a_j by a single service node with service rate $\mu_{a_j} = TR_{a_j}, j \in [\![W]\!]'$, transforming the original OQN (Figure 3) into a reduced OQN (Figure 5). We adopt the method proposed by Marchet et al. (2012) to solve the reduced OQN. The details presented in Appendix 6.

Finally, with the probability of a robot leaving aisle a_j after completing of the service at stopping point $s_{i,j}$, we can obtain the mean throughput rate of the aisle a_j, denoted as TR_{a_j}.

$$TT(\boldsymbol{x}) = \mu_c^{-1} + \sum_{l \in \mathcal{LS}} p_l \cdot (\overline{W}_l + \mu_l^{-1}) + \mu_{ca_0}^{-1} + \sum_{a_j \in \mathcal{A}} p_{a_j} \cdot \mu_{a_j}^{-1}, \tag{11}$$

where $\boldsymbol{x}$ represents an assignment scenario, that is defined as $\boldsymbol{x} = \{x_{u,v}\}$. Here, $x_{u,v}$ indicates whether destination u is assigned to drop-off point v (specified later). $\overline{W}_l$ represents the expected waiting time of a parcel at workstation $l, l \in \mathcal{LS}$ that is obtained following the solution of the reduced OQN.

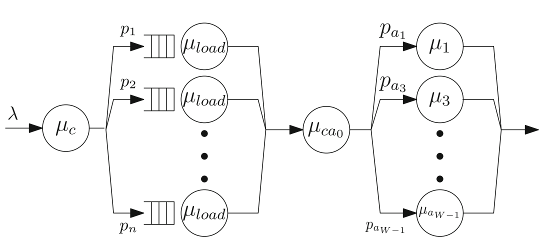


FIGURE 5 Reduced OQN of the RSS.

TABLE 2 Instances for simulation validation.

Instance	W	L	n	λ (per hour)
1	8	8	4	$U[10, 65]$
2	8	8	4	$U[15, 68]$
3	12	10	6	$U[10, 65]$
4	12	10	6	$U[10, 70]$
5	16	12	8	$U[5, 65]$
6	16	12	8	$U[10, 68]$

Note: System parameters come from our case company, including $w_a = 0.6m, w_d = 0.6m, v_m = 2m/s, t_{lu} = 2s, t_t = 1s, \underline{t}_{load} = 5, \overline{t}_{load} = 10$.

3.4 | Simulation validation of the OQN model

This section validates the accuracy of the analytical model of the OQN via simulation. The details of the simulation model are included in Appendix 9. Table 2 presents the system parameters and the instances. For each system configuration, we adopt two demand instances (Table 2), which can lead to the utilization of loading stations larger than 85% and 95%. Three validation criteria, including the throughput time TT, the expected waiting time of parcels at loading stations W_l, and the loading station utilization ρ_l, are considered. Absolute relative errors between analytical results and simulation results are used to measure the accuracy of the analytical model, denoted by $\delta_{TT}(\%)$, $\delta_{W_l}(\%)$ and $\delta_{\rho_l}(\%)$. For example, $\delta_{TT} = \frac{|TT_A - TT_S|}{TT_S} \times 100\%$, where TT_A and TT_S are the analytical and simulation throughput times, respectively.

Table 3 presents the mean simulation and analytical results and the absolute relative error. The average absolute relative error of throughput time TT is smaller than 3%, that of waiting time W_l is smaller than 4%, and that of loading station utilization ρ_l is smaller than 0.5%. Moreover, the distribution of the relative errors of TT, W_l, and ρ_l all fit an exponential-like distribution, with large errors occurring at a low frequency (see Figure A3 in Appendix 9). In conclusion, the analytical model of the OQN can provide accurate results for the RSS with not too low resource utilization and moderate variability of inter-arrival and service times.

4 | OPTIMIZATION OF DESTINATION ASSIGNMENT IN CONGESTED RSSS

In this section, we build an optimization model to minimize the throughput time of the RSS by assigning parcel destinations to DPs. The throughput time of the RSS is estimated using the OQN model developed in Section 3. To efficiently solve this optimization model, we design an ALNS algorithm.

4.1 | Optimization model of the destination assignment problem

To facilitate the formulation of the optimization model of the destination assignment problem, we first re-index DPs

TABLE 3 Validation results of the OQN model.

Instance	TT_S(s)	TT_A(s)	δ_{TT}(%)	W_l^S(s)	W_l^A(s)	δ_{W_l}(%)	ρ_l^S(%)	ρ_l^A(%)	δ_{ρ_l}(%)
1	45.10	45.14	1.20	19.36	19.41	2.79	88.05	88.06	0.31
2	84.01	83.88	2.30	56.39	57.08	3.27	95.60	95.59	0.40
3	49.97	49.98	0.34	15.94	15.96	1.08	85.30	85.30	0.36
4	89.08	88.86	2.36	54.64	54.53	3.85	95.34	95.32	0.31
5	59.78	59.51	0.51	15.74	15.66	1.34	85.69	85.64	0.32
6	98.34	98.63	1.65	54.38	54.85	3.10	95.20	95.23	0.33

Note: Mean values of TT, W, ρ and δ, derived from 30 replications of the simulation.

by setting $dp_u := d_{i,j}$, with $u = (j-1) \cdot W + i$, for $i \in [\![L]\!], j \in [\![W]\!]$. Then, we obtain a set comprising m DPs, where m is calculated as $W \times L$, and define this set as $\mathcal{DP} = \{dp_1, \dots, dp_m\}$. The set of destinations is defined as $\mathcal{DS} = \{dp_1, \dots, dp_{m'}\}$. We assume that the number of destinations m' equals that of the DPs m. Alternatively, we can define artificial zero-demand destinations or share low-demand destinations with a DP to make these numbers equal. A binary variable $x_{u,v}$ is used to determine whether the destination ds_v can be assigned to the DP dp_u. Then, $\boldsymbol{x} = \{x_{u,v} | u \in [\![m]\!], v \in [\![m']\!]\}$ is used to represent an assignment rule. Model (M.1) optimizes the destination assignment in the RSS.

$$\min TT(\boldsymbol{x}) \tag{M.1}$$

$$\text{s.t.} \begin{cases} \sum_{v \in [\![m']\!]} x_{u,v} = 1, \quad \forall u \in [\![m]\!] & (12) \\ \sum_{u \in [\![m]\!]} x_{u,v} = 1, \quad \forall v \in [\![m']\!] & (13) \\ x_{u,v} = 0, 1, u \in [\![m]\!], \quad v \in [\![m']\!]. & (14) \end{cases}$$

The objective function, that is, the throughput time $TT(\boldsymbol{x})$, tightly depends on the assignment rule of destinations to drop-off points $\boldsymbol{x}$. Constraint (12) ensure that each DP is used by one destination, and constraint (13) guarantee that each destination can only be assigned to one DP. To facilitate the description of the heuristic next, we use m to denote the number of destinations and DPs.

We then show that the destination assignment problem is an NP-hard problem by reducing it to a quadratic assignment problem. We decompose the objective function $TT(\boldsymbol{x})$ into two parts. One part is the robot traveling time, turning time, and dropping off time, which are linear with respect to the decision variables in the destination assignment problem. Another part is the robot congestion, which is nonlinear with respect to the decision variables. The difficulty of solving the problem relies on the nonlinear part, which mainly originates from the spillover effect of robot congestion in an aisle. To deal with this issue, we relax the original problem by only considering the robot congestion between any two adjacent DPs. In other words, we ignore the spillover effect of congestion and reduce the destination assignment problem into a quadratic assignment problem, which can be proven to be an NP-hard problem (Sahni and Gonzalez, 1976; Kerbachea & Smith, 1987). Therefore, we have Proposition 2 for the destination assignment problem.

Proposition 2. *The destination assignment problem is an NP-hard problem.*

Proof. Refer to Appendix 7. ■

4.2 | ALNS algorithm for the destination assignment problem

Based on the analysis of the destination assignment problem, we propose an *ALNS* algorithm to solve the model (M.1). ALNS is our method of choice because it has a great reputation of solving all kinds of scheduling problems, balancing quality with speed. For example, Turkeš et al. (2021) performed a meta-analysis of ALNS implementations. They showed that the ALNS can be used widely for a broad range of routing problems. Li et al. (2007) showed that "procedures based on adaptive large neighborhood search … performed well" on a large test of different VRP problems. Of course, the performance (quality and speed) depends on the tuning of the algorithm, both on the parameters and the selection of operators. The parameter settings in the paper by Ropke and Pisinger (2006) have been shown to be quite robust. The ALNS algorithm used in this paper follows the basic framework of Ropke and Pisinger (2006), but is modified according to our problem characteristics. The key differences between our ALNS algorithm and others are in both the initial solution generation and the neighborhood operators. All of them are designed to consider the impacts of travel distance, robot congestion, and their combination.

With a slight adjustment to the notation, we first denote a feasible assignment scenario as S_{ass} to indicate a set consisting of all DPs indexes and their corresponding assigned m destinations. Specifically, $S_{ass} = \{(u, v) | x_{u,v} = 1, u, v \in [\![m]\!]\}$, where (u, v) is an *index pair* that consists of a DP ds_u and its assigned destination ds_v.

In the first step of the algorithm, an initial feasible solution S_{ini} is created. Then, the global optimal solution, denoted as S_{glo}, and the current optimal solution, denoted as S_{cur}, are both initialized to S_{ini} and proceeds to the iteration part. In each iteration, the algorithm attempts to find a solution with a lower objective value, that is, the throughput time, by modifying the current solution using a destroy operator, denoted as $d(\cdot)$, and a repair operator, denoted as $r(\cdot)$. The definition of these operators is presented in Section 4.2.2. The selection of $d(\cdot)$ and $r(\cdot)$ is based on the adaptive mechanism that utilizes a roulette-wheel selection method. Additionally, the algorithm incorporates a simulated annealing method to accept suboptimal solutions with a certain probability. Finally, the global optimal solution S_{glo} is obtained and designated as an output. Appendix 8 presents the pseudocode of the ALNS algorithm. Appendix 11 gives the tuning process for the parameters of the ALNS algorithm.

4.2.1 | Initial solution

We calculate the throughput times of the HC, LC, and BA heuristic assignment rules, selecting the one yielding the minimum throughput time as the initial solution S_{ini}. The generation procedures of these assignment rules are detailed as follows:

- ***HC assignment rule***: Sort the destinations ds_v ($ds_v \in \mathcal{DS}$) in descending order according to their demands (the sorted destinations and the set of the sorted destinations are denoted as ds'_v and $\mathcal{DS}'$, respectively) and sequentially assign the sorted destinations ds'_v ($ds'_v \in \mathcal{DS}'$) to the DP dp_u ($dp_u \in \mathcal{DP}$).
- ***LC assignment rule***: Sort the destinations ds_v ($ds_v \in \mathcal{DS}$) in ascending order according to their demands (the sorted destinations and the set of the sorted destinations are denoted as ds'_v and $\mathcal{DS}'$, respectively) and sequentially assign the sorted destinations ds'_v ($ds'_v \in \mathcal{DS}'$) to the DP dp_u ($dp_u \in \mathcal{DP}$).
- ***BA rule***: This rule aims to achieve a balanced distribution of destination demands across drop-off points, and its generation procedure is described in Algorithm 1. The algorithm commences by sorting destination demands in ascending order. Subsequently, the algorithm partitions the sorted set into $\frac{m}{2}$ subsets, each of which pairs a low-demand destination with a high-demand destination. These subsets are then assigned to DPs in an alternating manner, ensuring a balanced distribution of destination demands among all DPs.

4.2.2 | Destroy and repair operators

During the search process of the ALNS algorithm, new feasible solutions are generated by destroying and repairing the current solution before the algorithm meets the terminating condition. The approaches to destroy and repair a solution are denoted as the destroy and repair operators, respectively. Considering the characteristics of the assignment problem investigated in this paper, we build four destroy operators and three repair operators, which are described as follows:

Destroy operators: A destroy operator $d(\cdot)$ selects a certain number of index pairs of the current solution S_{cur} and detaches the destinations of these pairs from their corresponding DPs. To diversify the search for a new solution, we introduce a certain level of randomness to destroy different index pairs of S_{cur}. The number of destroyed pairs in S_{cur} is controlled by $min(max(Q_{\min}, \sigma m), Q_{\max})$, where σ is a parameter to adjust the number of destroyed pairs in S_{cur}, and $Q_{\min}$ and $Q_{\max}$ define the minimum and maximum number of destroyed pairs, respectively. We set $Q_{\min} = 3$ given that the calculation process of the OQN in Section 3.2 needs at least three DPs, and we set $Q_{\max} = 20$ to avoid excessive calculation efforts. Four destroy operators are defined as follows:

- *Random destroy* $d(R)$: This operator randomly selects q destinations to be detached from their corresponding DPs. Specifically, q index pairs are randomly selected from the current solution S_{cur}, and the destinations of these pairs are detached from their corresponding DPs.
- *Worst destroy* $d(W)$: This destroy operator is adapted from the worst destroy operator, as proposed by Ropke and Pisinger (2006). The main idea is to select q index pairs from the current assignment scenario S_{cur}, which leads to the largest q changes in the throughput time when these pairs are destroyed. Then, we detach the destinations of these pairs from their corresponding DPs.
- *Congestion-based destroy* $d(C)$: This destroy operator is based on the congestion time (the calculation of the congestion time of the RSS is detailed in Appendix 3). Specifically, it selects q index pairs from the current assignment scenario S_{cur}, that leads to the largest q changes in the robot congestion time (calculated by Equation (A6) in Appendix 3) when these pairs are destroyed. Then, we detach the destinations of these pairs from their corresponding DPs.
- *Distance-based destroy* $d(D)$: This destroy operator is constructed from a travel distance perspective. It selects q index pairs from the current solution S_{cur}, which results in the largest q changes in robot traveling time (calculated by Equation (4)) when they are destroyed. Then, we detach the destinations of these pairs from their corresponding DPs.

Repair operators: A repair operator $r(\cdot)$ reassigns the detached destinations to the detached DPs, using the newly formed index pairs to generate a new solution S_{new}. To avoid building the same solution that is encountered several times during the search, a noise function is introduced, as suggested by Chen et al. (2021). The three repair operators are defined as follows:

- *Random repair* $r(R)$: This repair operator randomly reassigns a destroyed destination index to a destroyed DP index one by one until all destroyed indexes of destinations and DPs have been repaired.
- *Greedy repair* $r(S)$: This repair operator is adapted from the basic greedy repair operator as proposed by Ropke and Pisinger (2006). It reassigns a destroyed destination index to a destroyed DP index with the lowest repair cost until all destroyed indexes of destinations and DPs are repaired.

The repair costs are determined by the difference in the throughput time.

- *Sequentially greedy repair with a noise function* $r(SN)$: It is a variant of the previous operator but avoids the same solution to a certain extent. $r(SN)$ adds an additional value to change the least repair cost, thereby making it artificially nonoptimal. The additional value is calculated by $T_h \mu \epsilon$, where T_h is the largest throughput time, μ is a noise control parameter, and ϵ is a stochastic variable within the interval $[-1, 1]$.

4.2.3 | Adaptive mechanism

In this part, we introduce the adaptive mechanism to select a destroy operator and a repair operator in the iteration process. The selection of the above operators is based on the roulette-wheel procedure. Specifically, a destroy operator $d(\cdot)$ is selected based on its weight among all destroy operators in an iteration. We define the weight of $d(\cdot)$ in the iteration z as $\eta_z^-(d(\cdot))$, and calculate the probability of selecting $d(\cdot)$ by $\eta_z^-(d(\cdot))/\sum_{d(\cdot)}\eta_z^-(d(\cdot))$. Similarly, we define $\eta_z^+(r(\cdot))$ as the weight of the repair operator $r(\cdot)$ and calculate the probability of selecting $r(\cdot)$ in iteration z by $\eta_z^+(r(\cdot))/\sum_{r(\cdot)}\eta_z^+(r(\cdot))$. $\eta_z^-(d(\cdot))$ and $\eta_z^-(r(\cdot))$ are calculated according to Equation (15) and Equation (16), respectively.

$$\eta_z^-(d(\cdot)) = \omega \cdot \eta_{z-1}^-(d(\cdot)) + (1-\omega) \cdot s_z^-(d(\cdot))/c_z^-(d(\cdot)), \quad (15)$$

$$\eta_z^+(r(\cdot)) = \omega \cdot \eta_{z-1}^+(r(\cdot)) + (1-\omega) \cdot s_z^+(r(\cdot))/c_z^+(r(\cdot)), \quad (16)$$

where $s_z^-(d(\cdot))$ and $s_z^+(r(\cdot))$ represent the scores of the destroy operator $d(\cdot)$ and the repair operator $r(\cdot)$ from the beginning of the iteration procedure to iteration z, respectively. The determination of these scores is presented in Appendix 11. $c_z^-(d(\cdot))$ and $c_z^+(r(\cdot))$ indicate the number of times the destroy operator $d(\cdot)$ and the repair operator $r(\cdot)$ are selected, respectively, from the beginning of the iteration procedure to iteration z. To guarantee the applicability of Equation (15) or Equation (16), the initial values of $c_0^-(d(\cdot))$ and $c_0^+(d(\cdot))$ are set to unity. $\omega \in \{0, 1\}$ is used to adjust the influence of the destroy and repair operators in the current iteration.

4.2.4 | Acceptance criterion

After a new feasible solution is generated using the above operators, an acceptance criterion $Accept(\cdot,\cdot)$ is used to determine whether to accept the new feasible solution or not. $Accept(\cdot,\cdot)$ incorporates the simulated annealing mechanism introduced by Kirkpatrick et al. (1983) to accept both superior solutions and inferior solutions with a certain probability. If the quality of the new solution S_{new} is better than the current solution S_{cur}, then the new solution is accepted as the current solution, that is, $S_{cur} = S_{new}$. Otherwise, the worse solution can be accepted with a probability p_{accept}, which is calculated by $p_{accept} = e^{\frac{TT(S_{cur})-TT(S_{new})}{T}}$, where $TT(S_{cur})$ and $TT(S_{new})$ are the throughput times of the current solution and the new solution, respectively. The variable T represents the current temperature and is updated using the recursion equation $T := a \cdot T$, where the cooling rate a is a binary value within the set $\{0, 1\}$.

5 | NUMERICAL EXPERIMENTS

This section presents three groups of numerical experiments. First, we validate the throughput time performance and the computational efficiency of the ALNS algorithm by comparing it with Gurobi, a tabu search algorithm, and three assignment rules (HC, LC, and BA rules, including the HC rule used by the case company). Then, we investigate the effects of destination demand, system layout, and path topology on the assignment of parcel destinations to drop-off points and the throughput time. Finally, we evaluate the potential cost savings attributable to the implementation of the ALNS assignment rule.

5.1 | Performance verification of the ALNS algorithm

A series of experiments including comparisons with Gurobi, three heuristics assignment rules, and a tabu search algorithm used to solve another assignment problem in RSSs (Xu et al., 2022), are conducted to validate the performance of the ALNS algorithm in this section. We first generate 18 groups of instances, denoted as $I_1 - I_{18}$, as shown in Table 4. Each instance consists of two pieces of information: the system size (W and L) and the distribution used to generate the destination demands, that is, D. For each instance, we generate 100 sets of destination demands, submit them to the examined assignment rules, and calculate the average throughput times.

5.1.1 | Comparison between the ALNS algorithm and Gurobi

Model M.1 is solved by both the ALNS algorithm and the Gurobi solver (version 6.5.1). We use instances $I_1 - I_9$, as our

TABLE 4 Instances used for numerical experiments.

Instance	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9
W, L	3,2			4,4			6,4		
D (per hour)	$U[5, 15]$	$U[5, 35]$	$U[5, 65]$	$U[5, 15]$	$U[5, 35]$	$U[5, 65]$	$U[5, 15]$	$U[5, 35]$	$U[5, 65]$
Instance	I_{10}	I_{11}	I_{12}	I_{13}	I_{14}	I_{15}	I_{16}	I_{17}	I_{18}
W, L	8,8			10,12			12,16		
D (per hour)	$U[5, 15]$	$U[5, 35]$	$U[5, 65]$	$U[5, 15]$	$U[5, 35]$	$U[5, 65]$	$U[5, 15]$	$U[5, 35]$	$U[5, 65]$

TABLE 5 Computational performance comparison between the ALNS algorithm and Gurobi.

	Gurobi				ALNS			
Instance	$\overline{TT}(s)$	*Gap*(%)	*#Opt*	*CPU*(*s*)	$\overline{TT}(s)$	Gap_a(%)	*#Opt*	*CPU*(*s*)
I_1	15.44	1.12	98	40.03	15.46	0.13	81	1.05
I_2	17.11	2.21	95	47.33	17.14	0.18	73	1.41
I_3	20.14	1.88	96	56.84	20.16	0.10	76	2.55
I_4	18.33	25.43	43	368.34	18.34	0.05	83	1.53
I_5	21.88	29.33	37	377.29	21.88	0.00	95	1.81
I_6	25.44	29.06	36	383.71	25.44	0.00	97	2.44
I_7	22.03	53.23	6	1746.39	22.03	0.00	95	4.33
I_8	27.59	56.67	1	1787.39	27.12	0.00	100	4.93
I_9	38.23	59.32	0	1800	37.76	0.00	100	5.05

Note: *Gap* is the average percentage gap reported by the Gurobi. Gap_a is the average percentage gap of the ALNS algorithm solution to the Gurobi's solution, and is calculated as $Gap_a = 100\% \cdot \frac{ALNS-Gurobi}{Gurobi}$. *#Opt* represents the number of times of getting the optimal solution.

result shows that Gurobi cannot find the optimal solution for instances $I_{10}-I_{18}$. Table 5 shows the average throughput time ($\overline{TT}$) over the 100 sets of destination demands, the average optimality gap reported by the Gurobi solver in percentage (*Gap*) in which the optimality has not been proven, the average percentage gap of the ALNS solution to the Gurobi's solution (Gap_a) and the number of times of obtaining the optimal solution (*Opt*), and the average computing time (*CPU* in seconds). We set a running time limitation of half an hour on Gurobi solver. The results show that the performance of the Gurobi solver degrades significantly with an increase in the instance size. The ALNS algorithm takes little time (less than 11 s) to solve the problem in all instances, and its gap to the optimal solution is less than 0.18%.

5.1.2 | Comparison between ALNS and heuristic assignments

In this section, we compare the performance of the ALNS assignment rule with that of the HC, LC, and BA assignment rules. The generation processes of the above three assignment rules are defined in Section 4.2, in which we use them to construct the initial solution of the ALNS algorithm. The heat maps of the above three assignment rules and the ALNS assignment rule using the instance I_{18} is shown in Figure 6. The heat maps show the assignments of destinations to DPs in the RSS with 12 rows and 16 columns. Each small square corresponds to a DP, and the number in it represents the destination demand assigned to this DP.

We further compare the performance of the HC, LC, and BA assignment rules with the ALNS assignment rule in all instances. The results are summarized in Table 6. The average percentage gap (gap_{Rule}) of the throughput time between the HC, LC, and BA assignment rules and the ALNS assignment rule is calculated. The result shows that neither the HC nor the LC assignment rule is a good choice for the RSS in terms of the throughput time. The best result provides a throughput time 16.33% larger (on average) than that of the ALNS assignment rule. The BA assignment rule provides relatively better results than both the HC and LC assignment rules, albeit with an average 7.49% larger throughput time than that of the ALNS assignment rule. Although the HC rule can minimize the expected travel time of robots on the mezzanine, it leads to serious congestion at the entrance of each north-to-south aisle. The LC rule works in reverse. Both the BA rule and the ALNS rule attempt to disperse the high demands over the drop-off points, effectively balancing robot congestion and travel time.

Apart from the comparison between the ALNS algorithm and the above heuristic assignment rules, we also conduct numerical experiments to compare the performance of the ALNS algorithm with a tabu search algorithm. The tabu search algorithm is used for the assignment problem of arriving parcels to loading stations in RSSs (Xu et al., 2022). The results are presented in Appendix 10 for the sake of space conservation. The ALNS outperforms the tabu search algorithm by 2.3% on average in terms of the throughput time.

5.1.3 | A case study

We use the real demand dataset of the RSS in our case company to evaluate the effectiveness of the ALNS algorithm. The demand information is derived from 15 randomly chosen days in five months covering November 2018 to March 2019. We compare the throughput time of the ALNS assignment rule with that of the assignment rule used by the real system and present the average percentage gap (*gap*) in Table 7. The ALNS assignment rule can reduce the throughput time by 30% on average compared with the assignment rule applied by the real system.

5.2 | Analysis on destination demand, system layout, and path topology

Some key factors may affect the assignment of parcel destinations to DPs and the throughput time obtained by the ALNS algorithm. In this section, we conduct sensitivity analysis on three key factors: destination demand, system layout, and path topology. We use different distributions to

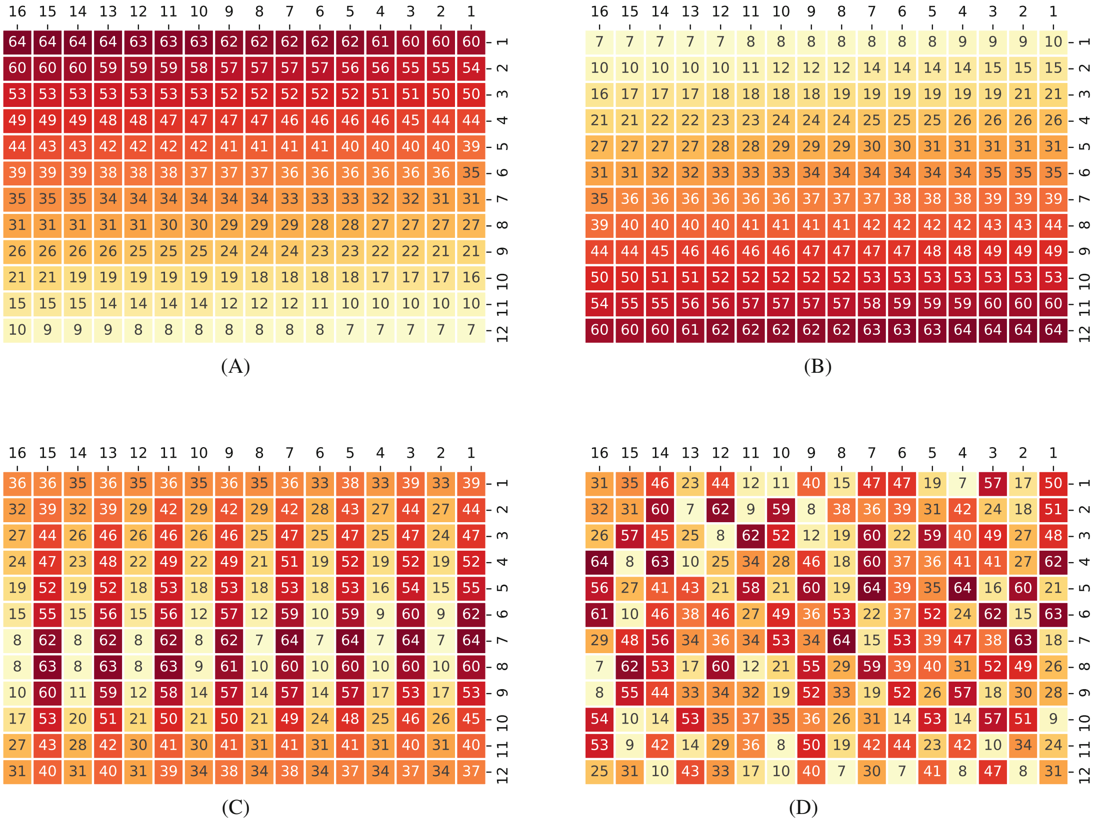


FIGURE 6 Heat maps of HC, LC, BA, and ALNS assignment rules. Input stations are located at the top.

TABLE 6 Performance comparison among HC, LC, BA, and ALNS assignment rules.

Instance	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9
gap_{HC}(%)	2.71	6.15	15.27	4.84	7.18	19.58	5.54	9.19	25.16
gap_{LC}(%)	8.76	8.27	11.92	9.48	10.33	15.30	8.80	12.38	18.82
gap_{BA}(%)	4.55	6.21	3.88	6.38	7.29	3.94	6.79	9.59	4.37
Instance	I_{10}	I_{11}	I_{12}	I_{13}	I_{14}	I_{15}	I_{16}	I_{17}	I_{18}
gap_{HC}(%)	8.24	10.32	36.33	9.42	12.31	42.21	10.42	13.77	55.21
gap_{LC}(%)	12.21	18.07	32.31	15.33	20.12	39.25	25.03	32.97	45.33
gap_{BA}(%)	9.32	10.75	4.63	10.21	12.51	4.87	11.36	13.21	4.97

Note: $gap_{Rule} = 100\% \cdot \frac{Rule-ALNS}{Rule}$.

TABLE 7 Evaluation results of ALNS algorithm using the demand dataset of our case company.

Date	11/5/18	11/11/18	11/23/18	12/2/18	12/10/18	12/24/18	1/3/9	1/11/19	1/19/19
gap(%)	50.12	27.13	30.24	38.84	17.87	19.98	35.56	38.59	37.77
Date	**2/15/19**	**2/18/19**	**2/20/19**	**2/27/19**	**3/5/19**	**3/10/19**	**3/20/19**	**3/29/19**	**3/31/19**
gap(%)	34.35	13.65	24.20	28.87	35.57	29.60	20.92	25.67	31.46
Avg(%)	30.02								

Note: $gap = 100\% \cdot \frac{RealSystem-ALNS}{RealSystem}$.

generate destination demands and examine how the variance of destination demands affects the ALNS assignment rule. Then, we vary the width-to-length ratio of the system with a given number of DPs to investigate the effect of the system layout on the system performance. Finally, we consider a path topology with dual lanes in each aisle and cross-aisle. Under these variations, we show how the assignment of destinations to DPs and the throughput time change.

5.2.1 | Destination demand

As mentioned earlier, the assignment of destinations to DPs balances robot congestion and robot travel distance. Which of these effects dominates depends on the variation among these demands. Our method determines how much robot congestion can be alleviated by using the ALNS algorithm. We conduct a sensitivity analysis to assess the impact of destination demand distributions. We consider a system with $W = 12, L = 12$, and 8 loading stations. Other parameters are derived from Table 2. The destination demands per hour are generated using three uniform distributions, namely, $U[5, 35]$, $U[10, 40]$, and $U[15, 45]$, corresponding to low, medium, and high demand levels, respectively. The assignments resulting from the ALNS algorithm are shown in Figure 7.

The HC assignment rule is close to the ALNS assignment rule in the low demand case. As demand increases, the ALNS assignment rule disperses the destinations with high demands, instead of grouping them close to the loading stations. For the system in which the destination demands are high, the robot congestion can be alleviated more by dispersing high-demand destinations over DPs. The ALNS algorithm finds a trade-off between robot congestion and robot travel distance by evenly assigning the destinations with high demands to the DPs in the middle of the mezzanine.

Insight: In the system in which the destination demands are low, the performance of the HC rule is close to the ALNS assignment rule. Focusing on minimizing the robot's travel distance is sufficient in this case. With the increase in destination demands, robot congestion becomes significant near drop-off points with large demands. The ALNS assignment rule finds the proper balance by arranging the large demands decentralized and further away from the loading stations.

5.2.2 | System layout

In the system examined in this paper, robot congestion mainly occurs in the north-to-south aisles. Whether a layout with short aisles leads to a smaller throughput time (or to a different destination assignment) than a layout with long aisles remains to be investigated. We use the width-to-length ratio $\frac{W}{L}$ to represent the layout shape of the RSS. We consider two system scales, namely, 64 DPs with 4 loading stations and 144 DPs with 8 loading stations, located on the north side of the DPs. Each system scale has three typical layouts: $\frac{W}{L} = \frac{1}{4}$ (a rectangular layout with long cross-aisles), $\frac{W}{L} = 1$ (a square layout), and $\frac{W}{L} = 4$ (a rectangular layout with long aisles). This configuration results in six system scenarios to be examined. For each destination, 100 sets of destination demands are generated (for each system layout scenario), using a uniform distribution of $U[5, 65]$ per hour. For each demand scenario, we use the ALNS algorithm to obtain the assignment of parcel destinations to DPs and the throughput time.

The throughput time of each system scenario over the 100 demand scenarios is shown in the box plots of Figure 8. Shortening the aisles benefits the throughput time, which leads to a layout with short aisles and long cross-aisles ($\frac{W}{L} = \frac{1}{4}$ in our case). Using a layout with long aisles (such as $\frac{W}{L} = 4$ in our case) leads to a much larger throughput time. However, a layout with too long cross-aisles (especially in a large system) may also hurt the throughput time, due to the long travel distance of the robot in the cross-aisles. For instance, the average throughput time in the instances with layout $\frac{W}{L} = \frac{1}{4}$ is larger than the average square layout $\left(\frac{W}{L} = 1\right)$ throughput time in the system with 144 DPs.

Then, we examine whether the layout affects the assignment of destinations to DPs obtained by the ALNS algorithm. To achieve this aim, we randomly choose one of the 100 demand scenarios obeying uniform distribution $[5, 65]$ per hour for each layout scenario and use the ALNS algorithm to obtain the destination assignment. The results of the system with 64 and 144 DPs are presented in Figures 9 and 10, respectively. The assignment of destinations to DPs under different layouts is similar.

Insight: The RSS should use a rectangular layout in which the cross-aisle may not be too long.

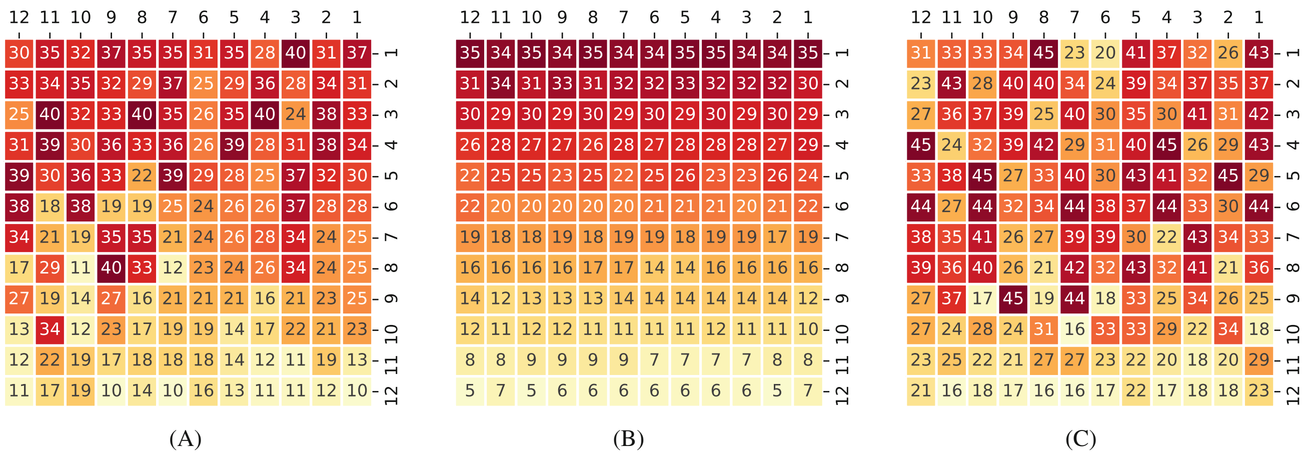


FIGURE 7 ALNS demand assignment for different demand levels.

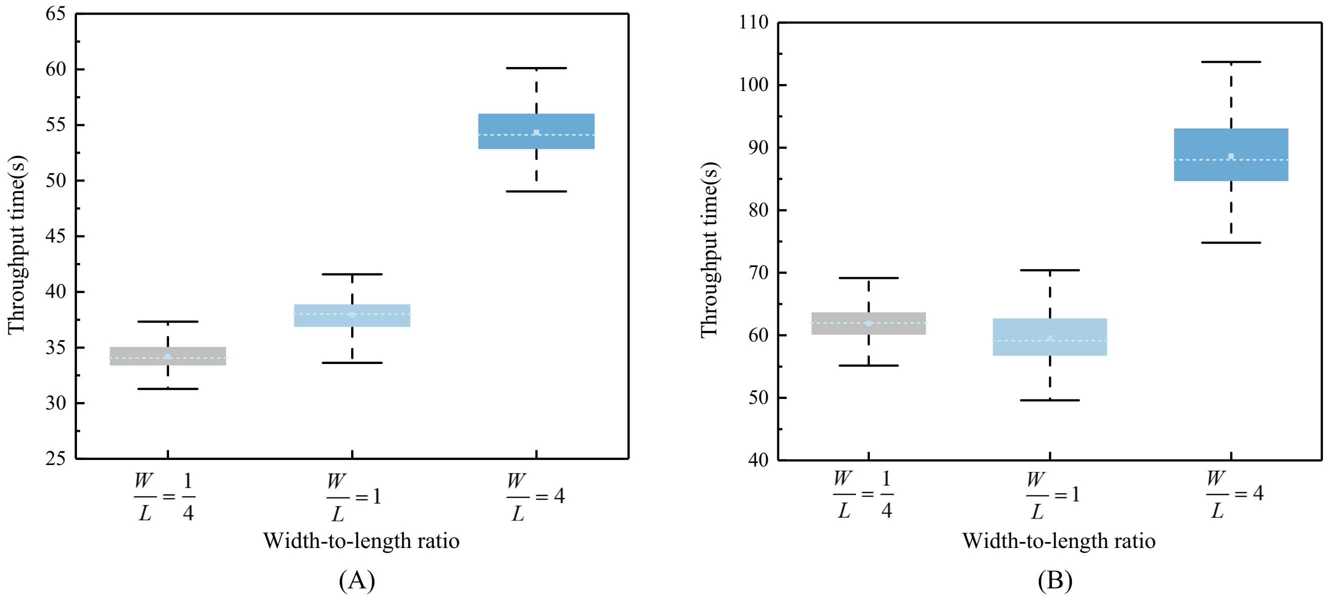


FIGURE 8 Effect of system layout on the throughput time of the RSS.

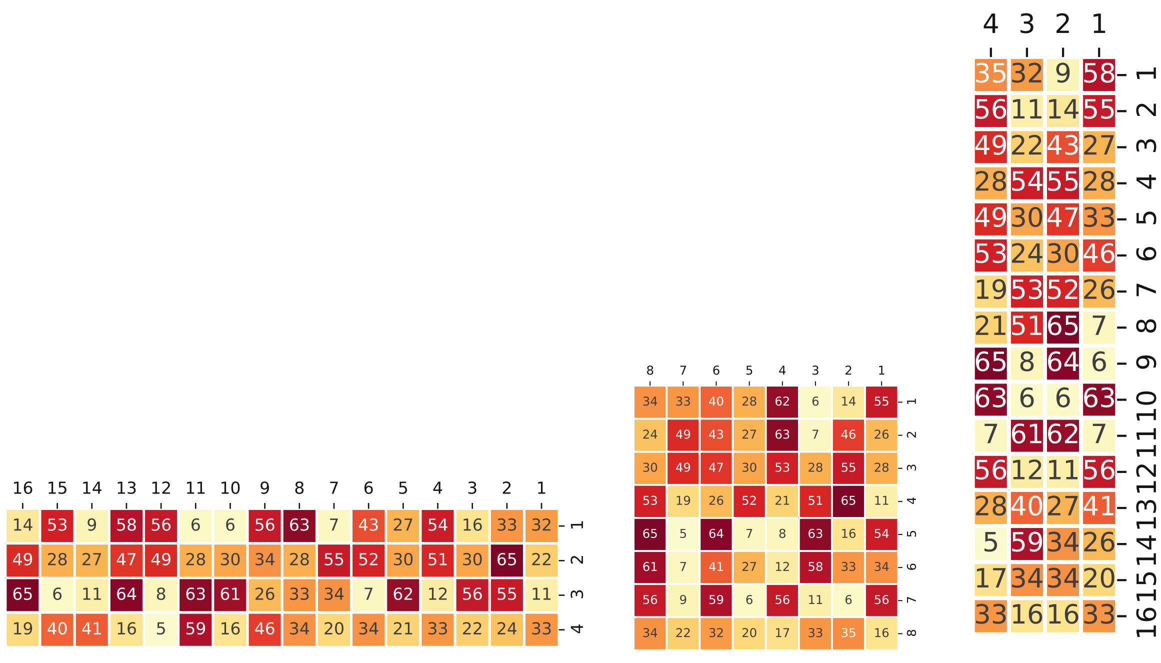


FIGURE 9 Effect of system layout on the assignment of destinations to DPs in the system with 64 DPs.

5.2.3 | Path topology

The layout in Section 3 assumes a travel path topology with one unidirectional lane in each aisle and cross-aisle. This typology leads to a compact footprint in which a robot may experience congestion while the travel distance is short. In this section, we consider a path topology with two unidirectional lanes in each aisle and a cross-aisle. This typology leads to a less compact layout, which may alleviate robot congestion. Our model can also handle this path topology by adjusting the aisle numbering a_i in Section 3.1, that is, changing $i = 1, 3, \dots, W-1$ to $i = 1, \dots, L$ given that the robot can release a parcel in each aisle. The robot will release the parcel in the north-to-south lane of an aisle, but it can directly return to the dwell area through the south-to-north lane in the aisle. The robot travel process in an aisle can also be modeled as a tandem queuing network like Figure 3, with a double space between two adjacent stopping points. Then, we consider the 18 system scenarios used in the performance validation of the ALNS algorithm (Section 5.1). The resulting average throughput time of both path topologies is shown in Table 8. The path topology with dual lanes outperforms the path topology with single lanes in terms of

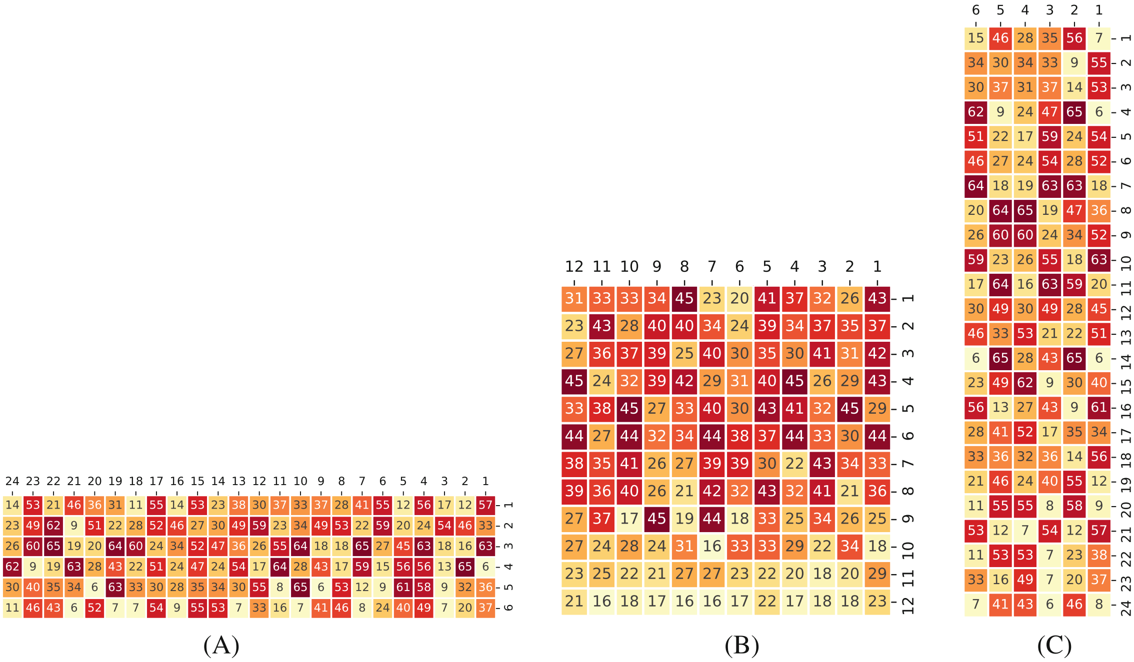


FIGURE 10 Effect of system layout on the assignment of destinations to DPs in the system with 144 DPs.

TABLE 8 Effect of path topology on the throughput time.

Instance	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9
$\overline{TT}_{single}(s)$	15.45	17.13	20.16	18.34	21.88	25.44	22.03	27.12	37.76
$\overline{TT}_{dual}(s)$	14.97	15.12	15.32	17.03	17.53	18.01	20.63	20.8	21.04
Instance	I_{10}	I_{11}	I_{12}	I_{13}	I_{14}	I_{15}	I_{16}	I_{17}	I_{18}
$\overline{TT}_{single}(s)$	32.48	38.18	51.29	43.64	51.09	65.53	59.52	70.08	95.31
$\overline{TT}_{dual}(s)$	25.48	25.75	26.07	33.29	34.86	35.87	43.52	46.26	49.31

the throughput time because robot congestion is alleviated. However, the dual-lane paths also require more floor space than the compact single-lane network.

The ALNS assignment rule following the single- and dual-lane layouts is shown in Figure 11A,B. We present one of the 100 randomly generated destination demands of instance I_{18}. The ALNS assignment rule tends to evenly disperse the destinations over DPs in the system with a compact layout, but it provides an assignment similar to the HC rule in dual-lane topology. However, when we increase the destination demands by generating them from the distribution $U[20, 80]$ per hour, the ALNS assignment rule tends to disperse the destinations with large demands and move them away from the loading stations (Figure 12).

Insight: A compact layout (with single-lane aisles and cross-aisles) leads to a larger throughput time than a dual-lane layout. Increasing the number of travel lanes in the aisles can mitigate the congestion, making the increase in the throughput time relatively smooth when the system size and destination demands increase. When the system faces large destination demands, dispersing the destinations with large demands among the DPs benefits throughput time.

5.3 | Cost analysis of the ALNS assignment rule

In this section, we evaluate the potential cost savings using the assignment rule obtained by the ALNS algorithm. Considering the impact of the system layout on the system cost, we compare the costs of the HC, LC, BA, and ALNS assignment rules under single- and dual-lane layouts. The system cost consists of the cost of loading stations, robots, and floor space. For each examined scenario, we optimize the system cost by determining the number of loading stations and robots used and the assignment rule of destinations to DPs, with a requirement on the throughput capacity (the maximum number of orders can be processed per unit time).

Subsequently, we transform the open queuing network into a closed queuing network to analyze the throughput capacity (TC) of the RSS with given resources (R robots, n loading stations, and the number of DPs in width W and length L). We first remove the arrival process and service process μ_c from Figure 3 and then set a network shortcut. Such a CQN can measure the throughput capacity of the RSS for a given number of resources. The detailed CQN and the solution method (approximate mean value analysis (Buitenhek et al., 2000)) are shown in Appendix 12. The objective is to minimize the

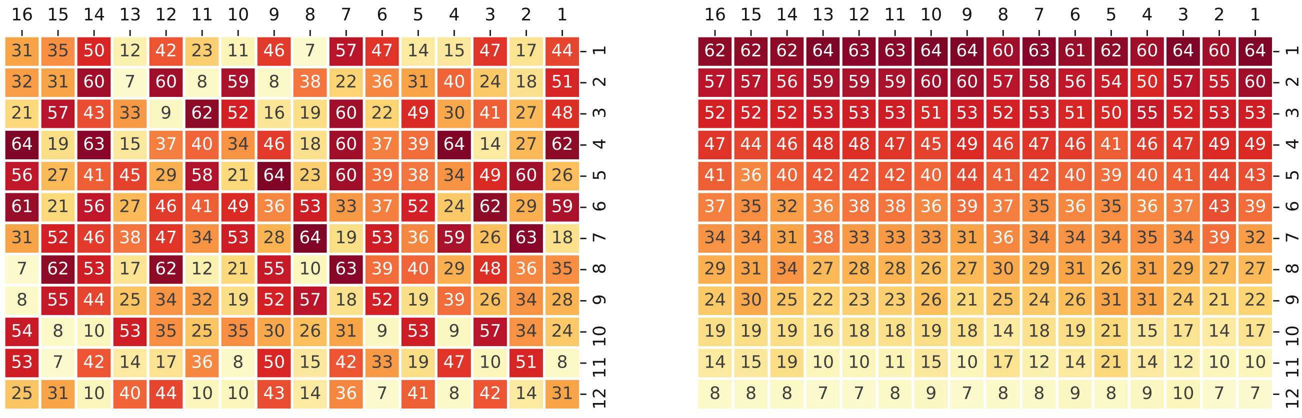


FIGURE 11 The ALNS assignment rule under different path topologies.

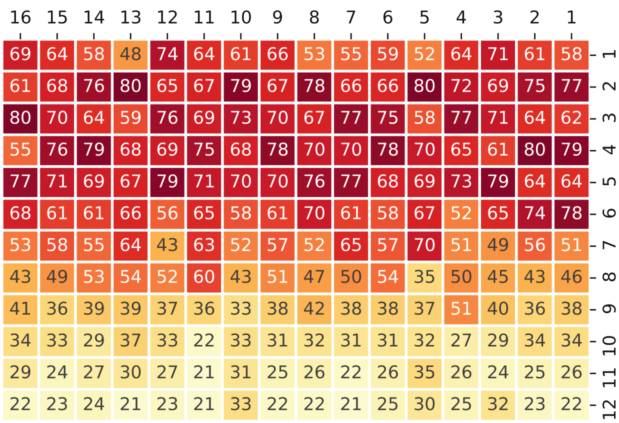


FIGURE 12 ALNS assignment rule in the system with a dual-lane topology and high demands.

total annual cost of the system with a requirement on the system throughput capacity ($TC \geq TC_{\min}$). This approach leads to Model M.2.

$$\min C(n, R, p) = C_{ls} \cdot n + C_R \cdot R + C_s \cdot A, \tag{M.2}$$

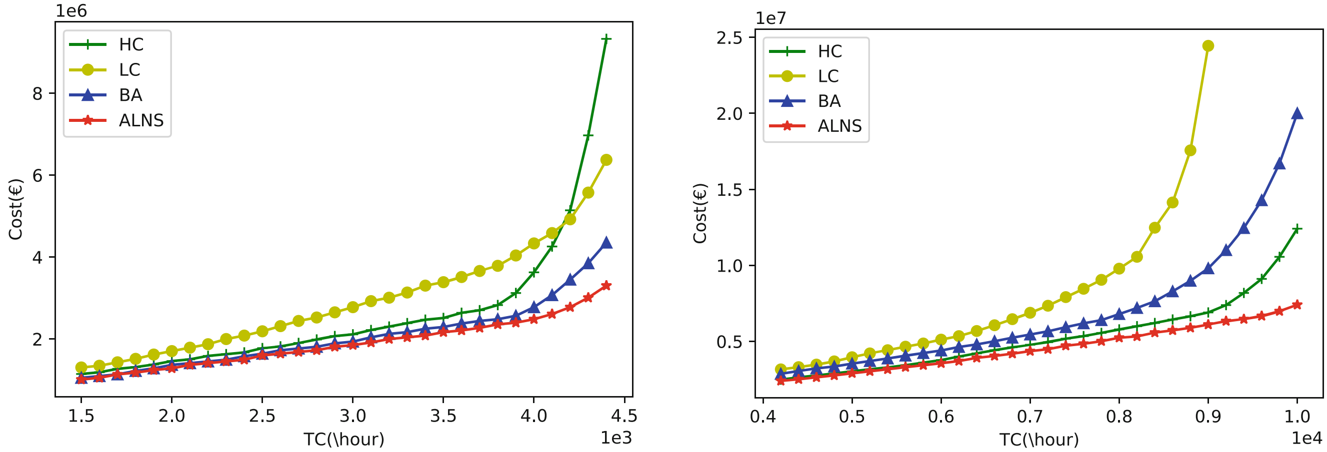
$$\text{s.t.} \begin{cases} TC \geq TC_{\min}, & (17) \\ n, R \in Z^+, & (18) \\ W, L \text{ are given}, & (19) \end{cases}$$

where C_{ls} is the annualized cost of a loading station, C_R is the annualized cost of a robot, and C_s is the annualized cost per square meter of space. A represents the area of the RSS. $A = W \cdot L \cdot (w_a + w_d)^2$ for the single-lane layout, and $A = W \cdot L \cdot (2 \cdot w_a + w_d)^2$ for the dual-lane layout. Constraint (17) guarantees the throughput capacity can meet the system requirements. The decision variables are given by n, R, and the assignment rule p, which is chosen among the HC, LC, BA, and ALNS assignment rules. The prices of the automated loading stations and robots are taken as €10 000 and €20 000, respectively. Moreover, the price of space per square meter of the RSS is €500 and annualized over 15 years. We consider a 10-year operating cycle and an annual interest rate of $IR = 5\%$. We obtain the annual cost of the loading station, robot, and space as follows:

$$C_{ls} = \sum_{t=1}^{10} \frac{10\,000(1+IR)^{t-1}}{10}, \quad C_R = \sum_{t=1}^{10} \frac{20\,000(1+IR)^{t-1}}{10},$$
$$C_s = \sum_{t=1}^{15} \frac{500(1+IR)^{t-1}}{15}. \tag{20}$$

Under each assignment rule, Model M.2 is solved by a greedy search procedure that enumerates all possible values of R and n to find a scenario with minimum cost and meets the requirement for throughput capacity. We take the system with $W = 12$ and $L = 16$ as an example for cost analysis and present the detailed results in Appendix 13 that consist of the optimal number of robots and loading stations and the optimal cost for each required TC. The comparative results of the HC, LC, BA, and ALNS assignment rules under single- and dual-lane topologies are shown in Figure 13.

As the throughput capacity increases, the cost of the system increases accordingly, given that more robots and more loading stations are required to achieve the required throughput capacity. However, the increased number of robots and loading stations depends on the assignment rule of destinations to DPs. The ALNS assignment rule provides a lower operating cost than the HC, LC, and BA assignment rules under both single- and dual-lane topologies by using fewer robots to achieve the required throughput capacity. The reduction of the robot fleet is caused by the alleviation of robot congestion and the reduction of robot travel distance. In the system using the single-lane topology, the congestion phenomenon at the entrance of the mezzanine becomes severe under the HC assignment rule, particularly when there is a high required throughput capacity. Increasing the number of robots has a little effect on improving the throughput capacity of the system in this case. In the system using the dual-lane topology, the travel time of robots is significantly prolonged under the LC assignment rule, necessitating a substantial increase in the number of robots to meet the required throughput capacity.



(A) (B)

FIGURE 13 Cost comparison among the HC, LC, BA, and ALNS assignment rules.

However, the cost associated with the BA assignment rule is lower than both the HC and LC assignment rules in the system with a single-lane topology, especially when there is a high throughput capacity requirement. A good explanation is that the BA assignment rule effectively balances robot congestion and robot travel time. Nonetheless, in the system using the dual-lane topology, the robot congestion is alleviated by increasing the number of lanes in aisles, and the BA assignment rule is worse than the HC assignment rule due to the larger robot travel time.

6 | CONCLUSION AND FUTURE WORK

This article investigates the assignment of destinations to DPs in a congested robotic sorting system, which deals with the trade-off between robot congestion and robot travel distance. We first build an OQN for throughput time estimation of the RSS in which each aisle is modeled as a tandem OQN with finite queue capacities. A decomposition method is designed for analyzing the throughput time in each aisle, of which convergence can only be guaranteed in the case of exponential distribution. On this basis, we replace the original OQN with a reduced OQN that can be solved by a product-form method. Then, we develop an integer programming model for the destination assignment problem and design an ALNS algorithm to obtain the solution. Numerical experiments are conducted to verify the effectiveness of the ALNS assignment rule in terms of both the throughput time and operating cost.

Then, we conduct sensitivity analysis on several key factors, including destination demands, system layouts, and path topologies. The results provide several insights for the practical operation of RSSs, especially considering robot congestion. The size of the destination demands affects the assignment. For small destination demands, using the HC assignment rule is preferred. For large overall destination demands, evenly distributing the destinations with the largest demands over the DPs is recommended. The system layout (width-to-length ratio) substantially affects the throughput time, but it only slightly affects the best assignments. Moreover, a larger system with dual-lane travel generally leads to a smaller throughput time than a system with a more compact single-lane travel layout. In this dual-lane travel layout, congestion can be mitigated to some extent. In cases in which congestion is only minor, the HC assignment rule performs well. In large-scale systems, the use of a proper assignment rule can lower the operating cost by using fewer loading stations and robots given the reduced robot congestion and robot travel distance.

For future studies, it may be interesting to consider the RSS with other layouts. An example is the T-sort system of Tompkins that uses a T-shape mezzanine in which the loading stations are arranged along the horizontal side and the drop-off points are arranged along the vertical side. In addition, it may be interesting to consider nonstationary demand and the randomness of the service process, under which dynamically reassigning destinations to drop-off points may further benefit the performance of the RSS. For such a study, researchers should consider that switching the roll containers on the bottom floor takes some time and can only be implemented at certain moments.

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CONFLICT OF INTEREST STATEMENT

No potential conflict of interest was reported by the authors.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from Deppon Express Co., Ltd. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the author(s) with the permission of Deppon Express Co., Ltd.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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